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| Score: / 10 |
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Calculus I (MATH 2200)

Spring 2020

Wednesday, 04/29/2020

Quiz 12: Section 5.1, 5.2

Name (Print): \_\_\_\_\_

Name (Print): \_\_\_\_\_

Name (Print): \_\_\_\_\_

## Section 5.1

5 pts

1. Write, but do not evaluate, the left Riemann sum for  $f(x) = x$  between  $a = 0$  and  $b = 2$  using  $n = 20$  subintervals. (Hint: For a left Riemann sum,  $x_k^* = a + (k - 1)\Delta x$ )

Ⓐ  $\sum_{k=1}^{20} (0.01k - 0.01)$

Ⓑ  $\sum_{k=1}^{20} (0.01 - 0.01k)$

Ⓒ  $\sum_{k=1}^{20} (0.02k - 0.02)$

Ⓓ  $\sum_{k=1}^{20} (0.01k^2 - 0.01k)$

With  $n = 20$ , the length of each subinterval is  $\Delta x = \frac{2 - 0}{20} = 0.1$ . The value  $x_k^*$  for the left Riemann sum is  $x_k^* = a + (k - 1)\Delta x = 0 + 0.1(k - 1) = 0.1k - 0.1$ . Thus,  $f(x_k^*) = 0.1k - 0.1$ , and the left Riemann sum is

$$\sum_{k=1}^{20} f(a + (k - 1)\Delta x) \Delta x = \sum_{k=1}^{20} (0.1k - 0.1)0.1 = \sum_{k=1}^{20} (0.01k - 0.01)$$

Section 5.2

5 pts

1. Assume that  $\int_0^5 f(x) \, dx = 3$  and  $\int_0^7 f(x) \, dx = -10$ . Evaluate  $\int_7^5 2f(x) \, dx$ .

(A) -14

(B) -26

(C) -17

(D) 26

$$\begin{aligned}\int_7^5 2f(x) \, dx &= -\int_5^7 2f(x) \, dx \\ &= -2 \int_5^7 f(x) \, dx \\ &= -2 \left( \int_0^7 f(x) \, dx - \int_0^5 f(x) \, dx \right) \\ &= -2(-10 - 3) \\ &= 26\end{aligned}$$